Harvard Recommendation Dossier: Yuri_Shelke

Referee Briefing Sheet & Draft LOR for Yuri_Shelke

Candidate:	Sham Satish Thakare (PRN: 72159719L, Seat: B191102032)
Recommender:	Yuri_Shelke, Distinguished Engineer, Capital One Systems Research
Programs Supported:	Harvard PhD (Math, CS, QSE) & Harvard SEAS ME (Fall 2027)
Deadlines:	SEAS ME: Dec 01, 2026 | PhD Programs: Dec 15, 2026
Submission Portal:	Harvard Griffin GSAS Slate Portal (link sent via automated email)

Candidate Academic & Research Summary

Dear Yuri_Shelke, Thank you for agreeing to provide a letter of recommendation for my applications to graduate programs at Harvard University. Below is an executive summary of key achievements, metrics, and milestones from our work together to assist you in drafting your letter.

• Mentorship Period: June 2024 to Present (2 years) as Pre-Doctoral Researcher at Capital One Systems Research.

• Core Systems Project: 'quorumshift' — dynamic quorum re-weighting Raft consensus engine implemented in modern C++20, eliminating p99 tail latency spikes under asymmetric cloud network degradation.

• Mathematical Consenus Proof: Derived formal topological invariants under bounded clock drift (Δ-bounded drift) guaranteeing majority intersection ($Q_i \cap Q_j \neq \varnothing$) and linearizability.

• Empirical Machine Learning: Falsified token predictive entropy as an uncertainty proxy in Group Relative Policy Optimization (GRPO) on GSM8K/MATH benchmarks, proving length collinearity ($r = +0.486$, $p < 10^{-6}$).

• Evaluation Frameworks: Formulated 'recovery_eval' and 'stateshift' for measuring counterfactual reasoning recovery in foundation models.

• Comparative Ranking: Ranks in the top 1% of young researchers supervised in systems maturity, theoretical depth, and research velocity.

Customizable Draft Letter of Recommendation

> *To the Members of the Graduate Admissions Committee, Harvard University: It is my distinct privilege to write this letter of highest recommendation for Sham Satish Thakare in support of his application for graduate study at Harvard University. As a Distinguished Engineer at Capital One leading systems architecture and distributed infrastructure research, I have had the privilege of mentoring and collaborating with hundreds of researchers and senior engineers over my career. I can state without hesitation that Sham ranks among the top 1% of young researchers I have ever supervised in terms of theoretical mathematical maturity, low-level systems execution, and autonomous research drive. I have mentored Sham closely for the past two years in his capacity as a Pre-Doctoral Researcher within our Systems Research group (June 2024 to Present). Sham joined our team after graduating with First Class with Distinction and the university Gold Medal in*

Mathematics from Savitribai Phule Pune University. What immediately distinguished Sham was his rare ability to bridge abstract mathematical proof with production-grade, bare-metal C++ systems architecture. During his pre-doctoral tenure, Sham took ownership of one of our most intellectually challenging research initiatives: resolving catastrophic tail latency in multi-region distributed databases under asymmetric network degradation. In conventional Raft and Paxos consensus deployments, network flapping or asymmetric partitions inevitably trigger chronic election loops and severe p99 latency spikes. Rather than settling for empirical heuristics, Sham approached this through formal topological invariants. He mathematically modeled quorum configurations as dynamic hypergraphs and derived a formal proof that under bounded clock drift (Δ-bounded drift), dynamic quorum re-weighting strictly preserves majority intersection invariants ($Q_i \cap Q_j \neq \emptyset$), guaranteeing linearizability and safety. He then personally implemented this engine, termed 'quorumshift', in modern C++20 using lock-free data structures and asynchronous I/O. In our rigorous chaos-engineering testbed, Sham's architecture completely eliminated leader oscillation and maintained deterministic write throughput throughout simulated cross-region network degradation. Simultaneously, Sham demonstrated remarkable empirical scholarship in foundation model reasoning geometry. Investigating the role of uncertainty in Reinforcement Learning with Verifiable Rewards (RLVR), Sham hypothesized that token predictive entropy in autoregressive generation is heavily confounded by sequence length. He designed and executed a 5-way matched ablation study across complex mathematical reasoning benchmarks (GSM8K and MATH), proving mathematically and empirically that prevailing heuristic uncertainty weighting acts merely as an unintended length penalty ($r = +0.486$, $p < 10^{-6}$). He went on to formulate the state-contrast continuation framework ('recovery_eval'), isolating genuine error recovery from unconditional generative capability. This work has resulted in multiple first-author manuscripts currently under peer review at IEEE Transactions and Elsevier venues. Beyond his technical prowess, Sham possesses an infectious curiosity, intellectual humility, and exceptional work ethic. He communicates complex mathematical arguments with lucidity and collaborates seamlessly with both research scientists and systems engineers. Whether in the PhD in Computer Science, PhD in Mathematics, PhD in Quantum Science and Engineering, or the Master of Engineering program at Harvard, Sham has my highest possible endorsement. He possesses the intellectual stamina, creativity, and foundational discipline to make seminal contributions to computer science and applied mathematics. I urge you to accept him into your incoming graduate class. Sincerely, Yuri Shelke Distinguished Engineer, Systems Research, Capital One